WEEK 1 · DAY 3 · INSTRUCTOR NOTES

Calm Down the Noise: Rolling Windows

For the trainer only.

The one idea for today: *a rolling window averages the last N rows and slides forward. It makes the real level visible, but it always costs something: a delay, a peek at the future (if centred), or hidden short events.*

At a glance

Length	~2 hours
Participants leave able to	smooth a series with `.rolling()`, explain why 7 days suits daily tickets, tell trailing from centred windows, use `.ewm()`, tell `.rolling` from `.resample`, and refuse to report an SLA from a smoothed line
Covers (original)	Week 3 Code Book: **B4 NOC tickets daily**, **B5 NOC tickets weekly**, and the mean-vs-minimum lesson from **A3 RF link quality**; Week 3 quiz topic *rolling windows and smoothing*
Datasets	`noc_tickets_daily.csv`, `noc_tickets_weekly.csv`, `rf_link_quality_hourly.csv` (all `Time-series/week-03/data/`)
Files you need open	animated slides (`1-slides/web-animated/`), your **solution** notebook, the rule book PDF

The day before: checklist

- ☐ Run your **solution** notebook top to bottom: all 14 checks should be ✅.
- ☐ Open the animated slides and let slide 6 (the sliding window) play once, so you know what it shows.
- ☐ Share the exercise link and the rule book PDF in the class group chat.

Run of show (2 hours)

Time	Block	What happens	Slides
0:00	**Hook**	Quiz warm-up from Day 2, then "Which day do faults rest?"	1–3
0:10	**The moving average**	Fuel-money analogy, the sliding window, the 2-year chart	4–7
0:25	**Choosing a window**	Why 7, then 3 vs 7 vs 30	8–9

Time	Block	What happens	Slides
0:35	**What smoothing costs**	Trailing vs centred, the lag, `.ewm`, the SLA trap, sparse counts, rolling vs resample, noise	10–17
1:00	**Hands-on**	Part A (15 min) then Part B (45 min). Walk the room.	18
1:45	**Wrap-up**	Scoreboards up, recap, three "why" questions	19
1:55	**Exit**	Share the glossary PDF. Tease Week 2.	–

Explain it like this (in plain English)

- **Rolling window:** take the last 7 days, average them, write the answer next to today. Tomorrow, drop the oldest day, add the newest, average again.
- **Why 7 for daily tickets:** every 7-day window contains exactly one Monday, one Tuesday … one Sunday. The quiet weekend is always inside, so it can't make the line jump. A 3-day window sometimes holds the weekend and sometimes doesn't, so it still zig-zags.
- **The first 6 NaN:** not a bug. On day 3 there are only 3 days to average, so a 7-day average genuinely doesn't exist yet.
- **Trailing vs centred:** trailing only looks back (safe, but late by about half the window). Centred looks back *and forward*. It lines up beautifully with events because it uses tomorrow's data. Fine for last year's report; never for live monitoring or forecasting. In Week 5 this mistake has a name: *leakage*.
- **`.ewm(alpha)`:** exponential smoothing. Recent days count more. High alpha follows the data and reacts fast; low alpha is calm and slow.
- **`.rolling(7)` vs `.resample("W")`:** rolling gives one value per day ("last 7 days as of today"); resample gives one value per calendar week ("total for that week"). Watch partial buckets at the edges.
- **The SLA rule:** an average spreads a 2-hour outage across 24 hours, and it disappears. SLA breaches are counted from raw readings.

Slide talking points

#	Say this	Ask this
3	Weekday 7.1 tickets, weekend 3.4: about half. Sunday is the quietest (3.1).	"Why are weekends quieter?"
5	Naira amounts are illustration only. The idea is what matters.	"Who checks their spending day by day? Week by week?"
6	Let the gold frame slide. The yellow number is the average inside the frame.	"Why is day 3 NaN?"
7	Grey is unreadable; blue shows the rainy-season rise.	"Which months are busiest?" (July to September)
8	3-day swing across the week: **2.79** tickets. 7-day swing: **0.02**.	"Why does 7 work?"

#	Say this	Ask this
11	Point at the red cells: days that haven't happened yet.	"Would you trust a forecast that used tomorrow's data?"
12	Trailing 30-day peak: 27 Aug 2024. Centred: 13 Aug 2024. **14 days** late (same gap in 2025).	"Bigger window: more or less delay?"
14	**12** real outage hours in August; **0** on the 24-hour average (it never drops below 4.55 dB).	"What would the customer say?"
15	P1: about 1.1 a day, 17% of days have none.	–
16	105 weekly buckets; the last one has only 2 days (9 tickets).	"Was that really a quiet week?"
17	Weekly mean **36.4** (the original code book says 38, which is wrong); typical swing ±5. Only 1 week in 52 is outside mean ± 2 swings: 47 tickets, in August.	"38 → 41: call a meeting?"

Answer key

Task	Answer	Common mistake → what to say
A1	`"date"`	Typing `"timestamp"` (that was Day 1's file)
A2	`"half"`	–
B1	`7`	`"7"` in quotes → `ValueError`
B2	`6` (`tk["roll7"].isna().sum()`)	Counting 7
B3	`3`	–
B4	`roll7_by_day.max() - roll7_by_day.min()` → **0.02**	Forgetting the brackets on `.max()`
B5	`True`	`"True"` with quotes
B6	`"no"` (the centred line is NaN on the last 15 days)	–
B7	`len(weekly)` → **105**	–
B8	`"high"`	Confusing "smooth" with "close": low alpha is smooth, high alpha is close
B9	`(tk["priority_1"] == 0).mean()` → **0.171**	Using `.sum()` (gives 125 days, not a share)
B10	`(margin < 0).sum()` → **12**, `(smooth < 0).sum()` → **0**	–
B11	`wk["tickets_raised"].mean()` → **36.42**, `"no"`	–

Task	Answer	Common mistake → what to say
🌶️	**1** (the week of 18 Aug 2025: 47 tickets)	–

Quiz link

The original Week 3 quiz (`Time-series/MCQ/week-03-mcq.csv`), topic **"rolling windows and smoothing"**, questions **13–18**, maps exactly to today:

Quiz question	Covered on slide
13 · Why seven days	8
14 · Trailing windows react late	12
15 · Centred windows are not for forecasts	11
16 · First rows are NaN	6
17 · Exponential smoothing alpha	13
18 · Never report SLAs from a smoothed line	14

Use them as a 5-minute warm-up at the start of Week 2 Day 1.

Troubleshooting

Problem	Fix
`ValueError: window must be an integer`	They typed `WINDOW = "7"`. Remove the quotes
`KeyError: 'roll7'`	The B1 cell after the blank wasn't run. Run it
B5 shows ❌ although they typed True	They wrote `"True"` in quotes
The rolling line looks identical to the raw line	They ran `rolling(1)` or forgot `.mean()`
Someone finishes early	🌶️ bonus, then: "try `rolling(7).std()` and tell me which season is more wobbly" (rainy season: about 2.6 vs 1.9 in Jan–Mar)

Notes on the original material

- The original code book says the weekly ticket mean is "about 38". The data says **36.4**. Use the real number.
- The original code book's day-of-week table is printed alphabetically (Friday, Monday, …). Our version orders Monday to Sunday.
- The original never shows code for rolling windows, centring or `.ewm`. That's all new here, written to match the quiz questions.